



MERN Stack MCQ Practice Questions

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JavaScript MCQs

Q1. What is the output?

```
console.log(typeof null);
```

- A) null
 - B) undefined
 - C) object
 - D) NaN
-

Q2. What is the output?

```
let a = [1, 2, 3];  
let b = a;  
b.push(4);  
console.log(a.length);
```

- A) 3
 - B) 4
 - C) undefined
 - D) Error
-

Q3. What is the output?

```
console.log(1 + "2" + 3);
```

- A) 6
 - B) 123
 - C) "123"
 - D) "33"
-

Q4. What is the output?

```
console.log([] == false);  
console.log([] === false);
```

- A) true, true
 - B) true, false
 - C) false, false
 - D) false, true
-

Q5. What is the output?

```
for (var i = 0; i < 3; i++) {  
  setTimeout(() => console.log(i), 0);  
}
```

- A) 0, 1, 2
 - B) 3, 3, 3
 - C) undefined, undefined, undefined
 - D) 0, 0, 0
-

Q6. Which is NOT a valid way to create a function?

- A) function foo() {}
 - B) const foo = function() {}
 - C) const foo = () => {}
 - D) function = foo() {}
-

Q7. What is the output?

```
const obj = { a: 1 };  
Object.freeze(obj);  
obj.a = 2;  
console.log(obj.a);
```

- A) 2
 - B) 1
 - C) undefined
 - D) Error
-

Q8. What is a closure?

- A) A function that is declared inside another function
 - B) A function that has access to variables from its outer scope even after the outer function returns
 - C) A way to close a browser window
 - D) A method to end a loop
-

Q9. What is the output?

```
console.log("5" - 3);  
console.log("5" + 3);
```

- A) 2, 8
 - B) "53", "53"
 - C) 2, "53"
 - D) NaN, "53"
-

Q10. What does `Array.prototype.reduce()` return?

- A) A new array
 - B) A boolean
 - C) A single accumulated value
 - D) undefined
-

Q11. What is the output?

```
const arr = [1, 2, 3];  
arr[10] = 11;  
console.log(arr.length);
```

- A) 3
 - B) 4
 - C) 10
 - D) 11
-

Q12. Which statement about `let` is FALSE?

- A) `let` is block-scoped
 - B) `let` can be redeclared in the same scope
 - C) `let` is not hoisted to the top like `var`
 - D) `let` creates a Temporal Dead Zone
-

Q13. What is the output?

```
Promise.resolve(1)  
  .then(x => x + 1)  
  .then(x => { throw new Error('Error!') })  
  .catch(() => 1)  
  .then(x => x + 1)  
  .then(x => console.log(x));
```

- A) 3
 - B) 2
 - C) Error
 - D) undefined
-

Q14. What is the output?

```
console.log(0.1 + 0.2 === 0.3);
```

- A) true
 - B) false
 - C) NaN
 - D) Error
-

Q15. What is the output of the event loop example?

```
console.log('1');  
setTimeout(() => console.log('2'), 0);  
Promise.resolve().then(() => console.log('3'));  
console.log('4');
```

- A) 1, 2, 3, 4
 - B) 1, 4, 2, 3
 - C) 1, 4, 3, 2
 - D) 1, 3, 4, 2
-

Q16. What does `===` check?

- A) Value only
 - B) Type only
 - C) Value and type
 - D) Reference only
-

Q17. What is the output?

```
const func = (a, b = a * 2) => a + b;  
console.log(func(5));
```

- A) NaN
 - B) 15
 - C) 10
 - D) undefined
-

Q18. Which method creates a new array?

- A) push()
 - B) pop()
 - C) map()
 - D) splice()
-

Q19. What is the output?

```
const person = { name: 'John' };  
const { name: myName } = person;  
console.log(myName);
```

- A) undefined
 - B) John
 - C) { name: 'John' }
 - D) Error
-

Q20. Arrow functions do NOT have:

- A) Parameters
 - B) Return statement
 - C) Their own `this`
 - D) Curly braces
-

Node.js MCQs

Q21. Node.js is:

- A) A programming language
 - B) A JavaScript runtime environment
 - C) A database
 - D) A web browser
-

Q22. Which module is used to handle file operations?

- A) `http`
 - B) `fs`
 - C) `path`
 - D) `url`
-

Q23. What is the purpose of `package.json` ?

- A) Store application data
 - B) Define project metadata and dependencies
 - C) Configure the database
 - D) Define routing
-

Q24. Which is TRUE about Node.js?

- A) It is multi-threaded by default
 - B) It uses synchronous I/O
 - C) It is single-threaded with event loop
 - D) It cannot handle concurrent requests
-

Q25. What does `npm install --save-dev` do?

- A) Installs package globally
- B) Adds package to `dependencies`

- C) Adds package to devDependencies
 - D) Removes the package
-

Q26. What is the output?

```
const path = require('path');  
console.log(path.extname('index.html'));
```

- A) index
 - B) .html
 - C) html
 - D) index.html
-

Q27. Which is used to read environment variables?

- A) process.env
 - B) global.env
 - C) require.env
 - D) module.env
-

Q28. What is middleware in the context of Node.js?

- A) Database connector
 - B) Functions that execute between request and response
 - C) A type of database
 - D) File system module
-

Q29. The require() function is used to:

- A) Export modules
 - B) Import modules
 - C) Create modules
 - D) Delete modules
-

Q30. Which method reads a file synchronously?

- A) fs.readFile()
 - B) fs.readFileSync()
 - C) fs.read()
 - D) fs.open()
-

Express.js MCQs

Q31. Express.js is a:

- A) Database
- B) Web framework for Node.js
- C) Programming language
- D) Template engine

Q32. Which method handles GET requests?

- A) `app.post()`
 - B) `app.get()`
 - C) `app.put()`
 - D) `app.fetch()`
-

Q33. What is `req.params` used for?

- A) Query string parameters
 - B) Route parameters
 - C) Request body
 - D) Request headers
-

Q34. What is the correct order of middleware execution?

- A) Random order
 - B) Alphabetical order
 - C) Order they are defined
 - D) Reverse order
-

Q35. Which middleware is used to parse JSON body?

- A) `express.static()`
 - B) `express.json()`
 - C) `express.urlencoded()`
 - D) `express.Router()`
-

Q36. How do you send a JSON response?

- A) `res.send()`
 - B) `res.json()`
 - C) `res.write()`
 - D) Both A and B
-

Q37. What status code indicates "Created"?

- A) 200
 - B) 201
 - C) 204
 - D) 301
-

Q38. In Express error-handling middleware, how many parameters are there?

- A) 2
 - B) 3
 - C) 4
 - D) 5
-

Q39. What does `next()` do in middleware?

- A) Ends the request
 - B) Passes control to the next middleware
 - C) Sends response
 - D) Throws an error
-

Q40. Which is used for modular routing?

- A) `express.static()`
 - B) `express.Router()`
 - C) `express.json()`
 - D) `express.urlencoded()`
-

MongoDB MCQs

Q41. MongoDB is a:

- A) Relational database
 - B) NoSQL document database
 - C) Graph database
 - D) Key-value store only
-

Q42. What is a Collection in MongoDB?

- A) Equivalent to a row in SQL
 - B) Equivalent to a table in SQL
 - C) Equivalent to a database in SQL
 - D) Equivalent to a column in SQL
-

Q43. What is the default primary key in MongoDB?

- A) `id`
 - B) `_id`
 - C) `key`
 - D) `primary_id`
-

Q44. Which operator is used for "greater than"?

- A) `$greater`
 - B) `$gt`
 - C) `$gte`
 - D) `$more`
-

Q45. What does `$push` do?

- A) Removes element from array
 - B) Adds element to array
 - C) Updates array element
 - D) Counts array elements
-

Q46. What is Mongoose?

- A) A MongoDB driver
 - B) An ODM (Object Data Modeling) library
 - C) A database
 - D) A query language
-

Q47. Which method finds one document and updates it?

- A) `updateOne()`
 - B) `findOneAndUpdate()`
 - C) `findAndModify()`
 - D) `updateById()`
-

Q48. What does `populate()` do in Mongoose?

- A) Adds new documents
 - B) Fills in referenced documents
 - C) Deletes documents
 - D) Counts documents
-

Q49. Which is used to define field validations in Mongoose?

- A) Model
 - B) Schema
 - C) Collection
 - D) Document
-

Q50. What does `$match` do in aggregation?

- A) Joins collections
 - B) Filters documents
 - C) Groups documents
 - D) Sorts documents
-

React.js MCQs

Q51. React is a:

- A) Framework
 - B) Library
 - C) Language
 - D) Database
-

Q52. What is JSX?

- A) JavaScript XML - syntax extension
 - B) A database query language
 - C) A CSS preprocessor
 - D) A testing framework
-

Q53. What is the Virtual DOM?

- A) The actual browser DOM
 - B) An in-memory representation of the real DOM
 - C) A JavaScript file
 - D) A CSS file
-

Q54. Which hook is used for state management?

- A) `useEffect`
 - B) `useState`
 - C) `useContext`
 - D) `useRef`
-

Q55. When does `useEffect` with empty dependency array run?

- A) On every render
 - B) Only on mount
 - C) Never
 - D) On unmount only
-

Q56. What is the correct way to update state?

```
const [count, setCount] = useState(0);
```

- A) `count = 1`
 - B) `count++`
 - C) `setCount(1)`
 - D) `this.state.count = 1`
-

Q57. Keys in React lists should be:

- A) Random
 - B) Index always
 - C) Unique among siblings
 - D) Global unique
-

Q58. What does `useRef` return?

- A) State value
 - B) Mutable ref object with `.current`
 - C) Context value
 - D) Callback function
-

Q59. Which hook is similar to Redux?

- A) `useState`
- B) `useReducer`

- C) `useEffect`
 - D) `useMemo`
-

Q60. What does `useMemo` do?

- A) Memoizes component
 - B) Memoizes computed value
 - C) Memoizes callback function
 - D) Creates ref
-

Q61. What does `useCallback` do?

- A) Memoizes component
 - B) Memoizes computed value
 - C) Memoizes callback function
 - D) Creates ref
-

Q62. Props in React are:

- A) Mutable
 - B) Read-only
 - C) Global
 - D) Optional always
-

Q63. What is prop drilling?

- A) Creating props
 - B) Passing props through many layers
 - C) Deleting props
 - D) Validating props
-

Q64. Which is used to avoid prop drilling?

- A) `useState`
 - B) `useContext`
 - C) `useRef`
 - D) `useEffect`
-

Q65. Controlled components have:

- A) No state
 - B) State managed by DOM
 - C) State managed by React
 - D) No props
-

Answers & Explanations

Q#	Answer	Explanation
1	C	<code>typeof null</code> returns "object" - this is a known JavaScript bug

2	B	Arrays are reference types; <code>b</code> points to same array as <code>a</code>
3	C	String concatenation: <code>1 + "2" = "12"</code> , then <code>"12" + 3 = "123"</code>
4	B	<code>==</code> does type coercion (true), <code>===</code> checks type too (false)
5	B	<code>var</code> is function-scoped; by the time callbacks run, <code>i</code> is 3
6	D	Invalid syntax
7	B	<code>Object.freeze()</code> prevents modifications
8	B	Closure = function + its lexical environment
9	C	- coerces to number, + concatenates strings
10	C	<code>reduce()</code> returns single accumulated value
11	D	Sparse array; length = highest index + 1
12	B	<code>let</code> cannot be redeclared in the same scope
13	B	<code>catch</code> returns 1, then <code>1 + 1 = 2</code>
14	B	Floating-point precision issue
15	C	Sync first, then microtasks (Promise), then macrotasks (<code>setTimeout</code>)
16	C	Strict equality checks both value and type
17	B	Default param <code>b = 5 * 2 = 10</code> , so <code>5 + 10 = 15</code>
18	C	<code>map()</code> returns new array; others mutate original
19	B	Destructuring with rename
20	C	Arrow functions don't have their own <code>this</code>
21	B	Node.js is a JavaScript runtime built on V8
22	B	<code>fs</code> module handles file system operations
23	B	<code>package.json</code> defines metadata and dependencies
24	C	Node.js is single-threaded with event loop
25	C	<code>--save-dev</code> adds to <code>devDependencies</code>
26	B	<code>path.extname()</code> returns extension including dot
27	A	<code>process.env</code> contains environment variables
28	B	Middleware = functions between request and response
29	B	<code>require()</code> imports modules
30	B	<code>readFileSync</code> is synchronous
31	B	Express is a web framework for Node.js

32	B	<code>app.get()</code> handles GET requests
33	B	<code>req.params</code> contains route parameters
34	C	Middleware executes in order of definition
35	B	<code>express.json()</code> parses JSON body
36	D	Both <code>send()</code> and <code>json()</code> can send JSON
37	B	201 = Created
38	C	Error middleware has 4 params: <code>err</code> , <code>req</code> , <code>res</code> , <code>next</code>
39	B	<code>next()</code> passes control to next middleware
40	B	<code>express.Router()</code> for modular routing
41	B	MongoDB is a NoSQL document database
42	B	Collection = Table
43	B	<code>_id</code> is the default primary key
44	B	<code>\$gt</code> = greater than
45	B	<code>\$push</code> adds element to array
46	B	Mongoose is an ODM for MongoDB
47	B	<code>findOneAndUpdate()</code> finds and updates
48	B	<code>populate()</code> fills in referenced documents
49	B	Schema defines validations
50	B	<code>\$match</code> filters documents in aggregation
51	B	React is a library (not framework)
52	A	JSX = JavaScript XML syntax extension
53	B	Virtual DOM is in-memory representation
54	B	<code>useState</code> for state management
55	B	Empty deps array = runs only on mount
56	C	Use setter function <code>setCount()</code>
57	C	Keys must be unique among siblings
58	B	<code>useRef</code> returns object with <code>.current</code>
59	B	<code>useReducer</code> is similar to Redux pattern
60	B	<code>useMemo</code> memoizes computed value
61	C	<code>useCallback</code> memoizes callback function

62	B	Props are read-only
63	B	Prop drilling = passing through many layers
64	B	useContext avoids prop drilling
65	C	Controlled = state managed by React

Score yourself:

- **55-65:** Excellent! Ready for interview 🎉
- **45-54:** Good! Review weak areas
- **35-44:** Fair. More practice needed
- **Below 35:** Focus on fundamentals first